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(PDEU)
Software Engineering(24CS209P)

Lab Experiment - 6

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Objective:

Understand how software components are organized and deployed.

Basic Theory:

Component diagram shows software parts.

Deployment diagram shows where software runs.

Lab Activities:

1. Choose architecture type
2. Draw component diagram
3. Draw deployment diagram

Deliverables:

1. Short architecture description:

The system architecture is based on the Model-View-Controller (MVC) pattern.

- **Model:** Represents the core data and business logic of the system. It includes classes such as Book, Student, Transaction, and User, and handles operations like book availability, borrowing records, and fine calculation.
- **View:** Represents the user interface of the system, developed using HTML/CSS (and optionally JavaScript). It allows users to interact with the system by searching books, viewing details, and performing actions like borrow or return.
- **Controller:** Acts as an intermediary between the View and Model. It processes user inputs (e.g., login, search, borrow requests), invokes appropriate Model operations, and updates the View with the results.

2. Component diagram:

- **Components:** Authentication Module, Record Management Module, Book manager, Transaction Manager, Borrow Book Module.
- **Database:** Mongo DB.

